

F O R T I M A N A G E R N S E 6

Lab 09 — Global Database, FortiGuard and Firmware Templates

01. The Global Database

Every lab so far has worked inside one ADOM. Policy, objects and packages are ADOM-scoped, so an organisation running several ADOMs — by region, by business unit, by tenant — has no way to enforce a rule across all of them. The Global Database is that mechanism.

It is a special ADOM that holds no devices. What it holds is policy that is pushed down into other ADOMs: header policies that sit above every local policy, footer policies that sit below them, and the objects those policies reference.



Security Fabric (0)					
Central Management (7)					
☐	root	FortiGate 7.4	VPN, FortiAP, FortiSwitch	1 Device (including 1 VDOM)	US/Policy (4700 PBT)
☐	FortiProxy	FortiProxy 7.4	VPN, FortiAP, FortiSwitch		US/Policy (4700 PBT)
☐	FortiRevalCenter	FortiRevalCenter 6.2	VPN, FortiAP, FortiSwitch		US/Policy (4700 PBT)
☐	FortiReval	FortiReval 6.2	VPN, FortiAP, FortiSwitch		US/Policy (4700 PBT)
☐	FortiCenter	FortiCenter 7.4	VPN, FortiAP, FortiSwitch		US/Policy (4700 PBT)
☐	7_ADOM	FortiGate 7.2	VPN, FortiAP, FortiSwitch		US/Policy (4700 PBT)
☐	Global Database	Global 7.4	VPN, FortiAP, FortiSwitch		US/Policy (4700 PBT)

Figure 1 — The ADOM list — Global Database alongside the per-product ADOMs.



☒	Global Database	Global 7.4	VPN, FortiAP, FortiSwitch	US/Policy (4700 PBT)
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Figure 2 — Global Database, running at Global 7.4.

► 1.1 How it fits together

The diagram below is the shape of the whole lab. It is worth reading before the screenshots, because the ordering it shows is what makes header and footer policies useful.

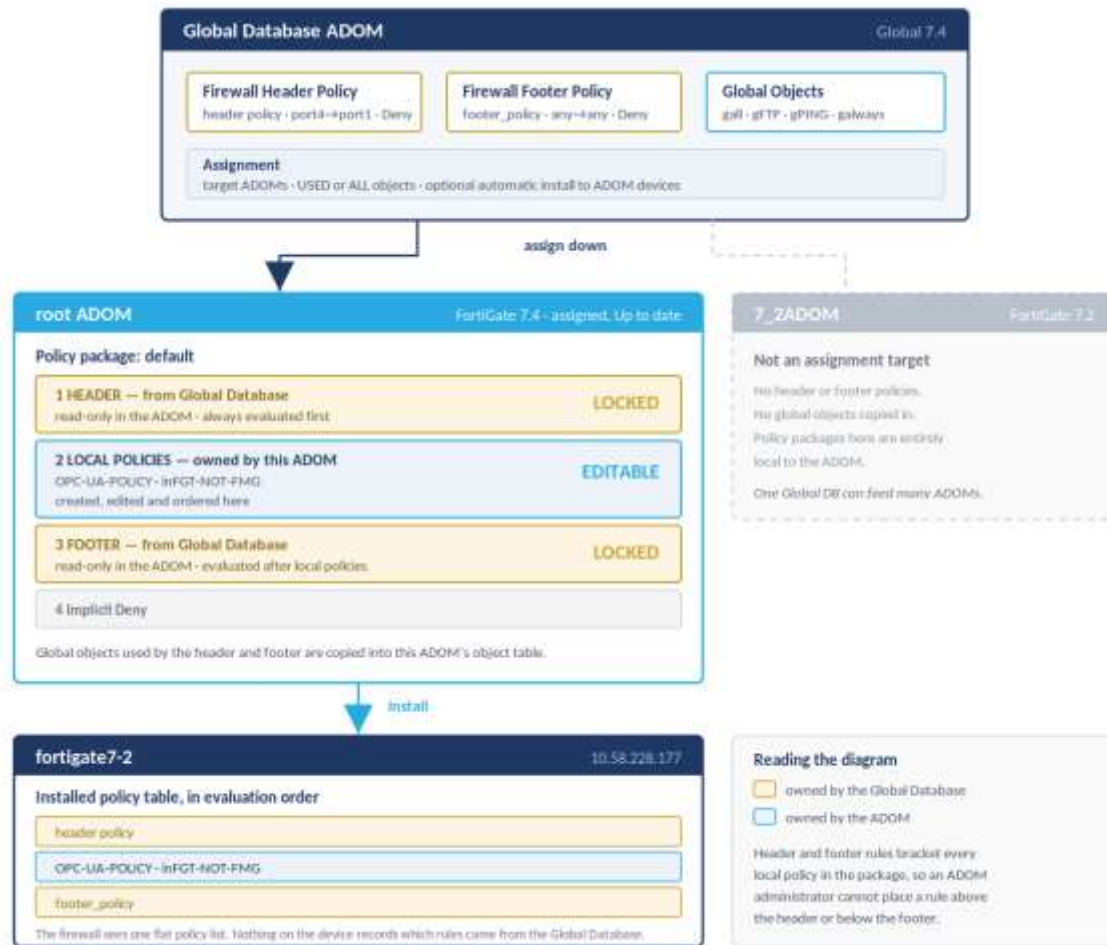


Figure 3 — The Global Database model — global policy assigned down into an ADOM, bracketing the ADOM's own policies, and installed to the device as one flat list.

The point is the bracket, not the sharing

A header policy is always evaluated before any policy the ADOM administrator can write, and a footer policy is always evaluated after all of them.

That ordering is what a central security team needs: a deny rule nobody downstream can place a permit above, and a catch-all nobody downstream can place a permit below. Sharing objects across ADOMs is a side effect, not the purpose.

► 1.2 Entering the Global Database

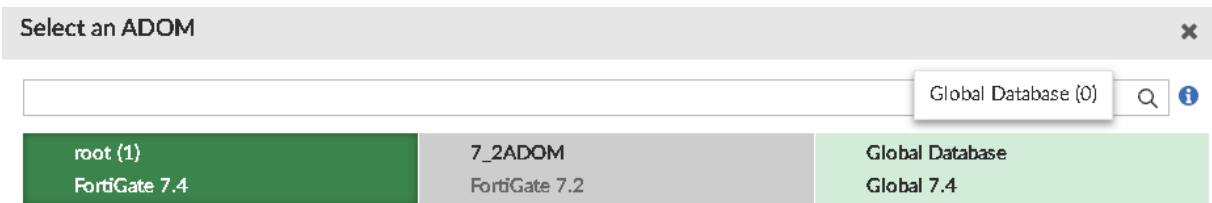


Figure 4 — Select an ADOM — Global Database listed with root and 7_2ADOM.

It behaves like any other ADOM in the interface, with one difference visible immediately: its policy package tree carries Firewall Header Policy and Firewall Footer Policy instead of a single Firewall Policy, and an Assignment entry in place of installation targets.

02. Header and Footer Policies

► 2.1 The header policy

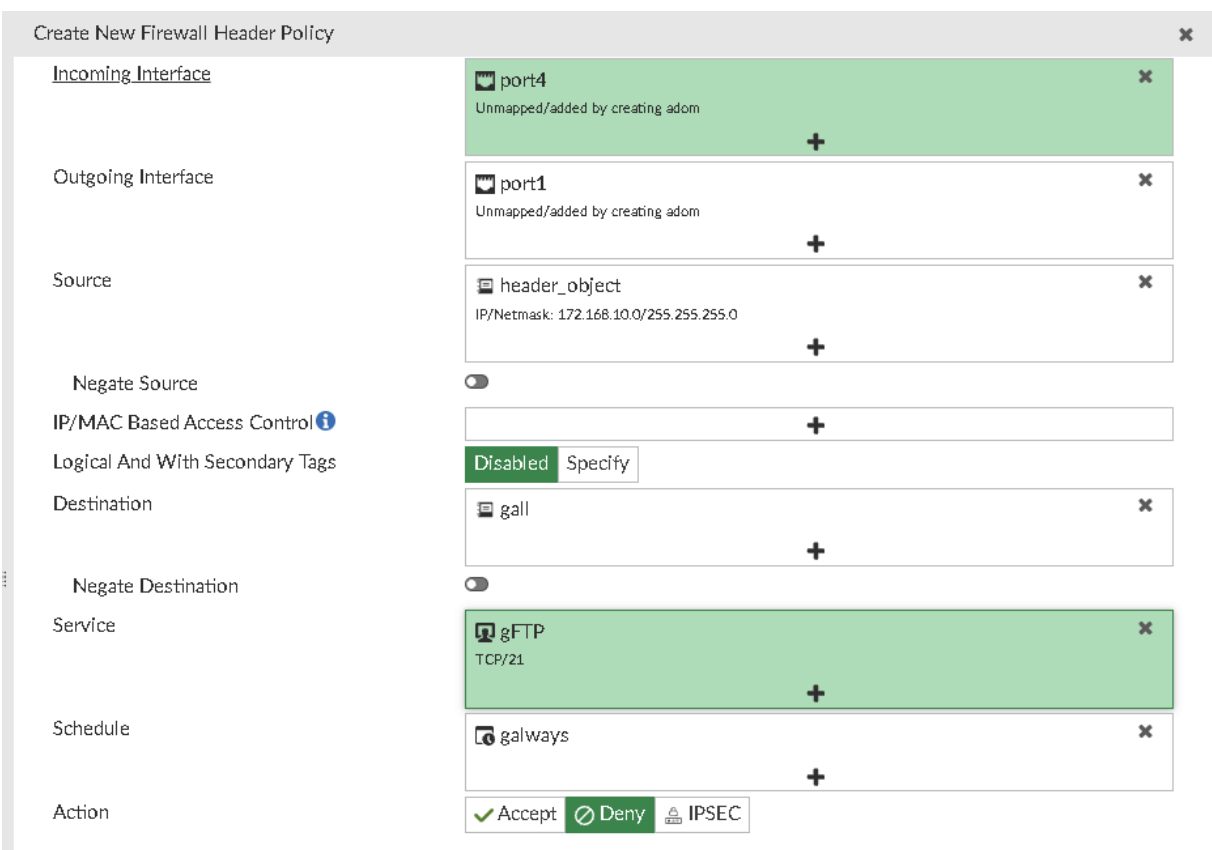


Figure 5 — Create New Firewall Header Policy, using global objects.

The objects are global ones created in this ADOM — header_object, gall, gFTP, galways. The naming convention is worth copying: a leading g makes it obvious in a downstream ADOM that an object arrived from the Global Database rather than being created locally.

Observation	The incoming and outgoing interfaces show port4 and port1 with the note "Unmapped/added by creating adom".
Why	The Global Database has no devices, so it has no real interfaces. These are normalized interface names created on the fly, and each receiving ADOM maps them to its own devices' physical ports.
Risk	A global policy pinned to port4 only matches on devices where port4 is mapped. Global header and footer policies are usually written against any, or against normalized interfaces that exist in every target ADOM.



Figure 6 — The header policy in the Global Database package.

▸ 2.2 The footer policy

Figure 7 shows the configuration for a new Firewall Footer Policy. The settings are as follows:

Field	Value
ID	200
Name	footer_policy
Type	Standard
Incoming Interface	any
Outgoing Interface	any
Source	gall
Negate Source	disabled
IP/MAC Based Access Control	Disabled
Logical And With Secondary Tags	Specify
Destination	gall
Negate Destination	disabled
Service	gPING
Schedule	galways
Action	Accept, Deny, IPSEC

Figure 7 — Create New Firewall Footer Policy — any to any, gall to gall, gPING, Deny.

ID	Name	From	To	Source	Destination	Schedule	Service	Action	Secure
200	footer_policy	any	any	gall	gall	galways	gPING	Deny	

Figure 8 — The footer policy in the package.

Both policies deny. That is the normal shape for global policy: the header denies specific traffic nobody downstream may permit, and the footer denies a broad class of traffic nobody downstream may reach past. A global policy that accepts is rarer, because it grants access across every ADOM it lands in.

03. Assigning to ADOMs

A global package does nothing until it is assigned. Assignment is the Global Database equivalent of installation targets, except the target is an ADOM rather than a device.



Figure 9 — Add Assignment Target — selecting the root ADOM and the policy packages within it.

Choice	What it does
All Policy Packages	Every package in the target ADOM receives the header and footer policies, including packages created later.
All Policy Packages except Specified ones	Blanket coverage with named exemptions — the usual shape when one package legitimately cannot carry the global rules.
Specify Policy Packages To Assign	Only the named packages receive them. New packages are not covered until they are added here.

► 3.1 Object scope and automatic install



Figure 10 — Assign Global Policy Package — object scope and the automatic install toggle.

Setting	Choice made	What it means
Assign USED Objects Only	Selected	Only the global objects referenced by the header and footer policies are copied into the target ADOM.
Assign ALL Objects	—	The entire global object table is copied down, whether referenced or not. Useful for standardising an object library, at the cost of filling every ADOM with objects it may never use.
Automatically Install Policies to ADOM Devices	Off	With it off, assignment updates the ADOM's packages and stops. Devices receive the global rules at the next install. With it on, FortiManager installs to every device in the target ADOM as part of the assignment.

The automatic install toggle is the one to think about

Leaving it off means the global rule exists in the ADOM but is not yet enforced on any firewall — enforcement waits for whoever next installs that package.

Turning it on pushes a central security decision straight to every device in the ADOM, bypassing the local install preview. Both are defensible; the mistake is not knowing which one is set.

Assign Global Policy Package - default

5%

Total: 0/1, Pending: 1, In Progress: 0, Completed: 0

Show Details

View Progress Report

Search...

#	Name	Time Used	Status
1	root[assign]	N/A	

Figure 11 — The assignment running.



Figure 12 — Assignment complete — root reported as Up to date.

04. What the Receiving ADOM Sees

Switching back to the root ADOM, every policy package now carries the global rules.

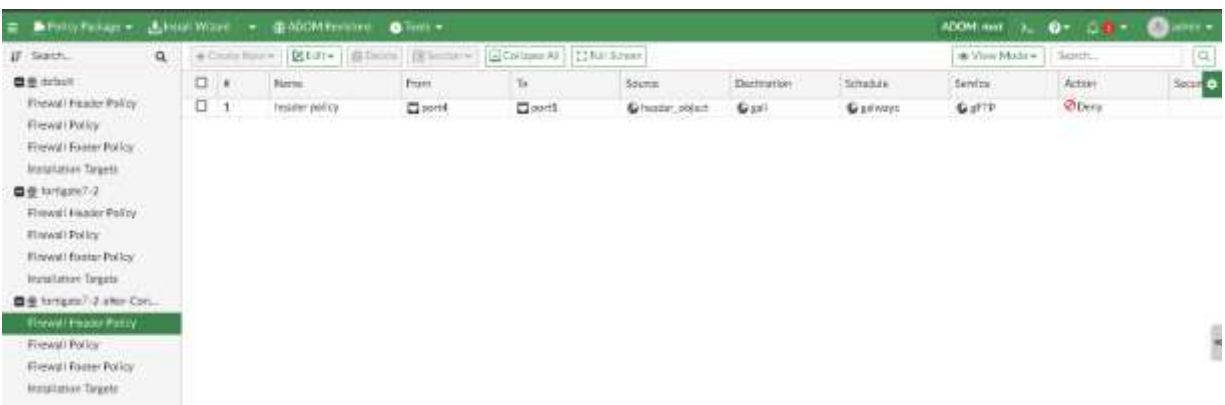


Figure 13 — The root ADOM — Firewall Header Policy present in each package alongside the local Firewall Policy.

All three packages built in Labs 07 and 08 — default, fortigate7-2 and fortigate7-2-after-Conflict — received the header and footer, because the assignment was made with All Policy Packages.

▸ 4.1 Read-only in the receiving ADOM

Opening the header policy in the root ADOM shows it in a View dialog rather than an Edit one. A service can be selected in the picker, and the change cannot be saved.

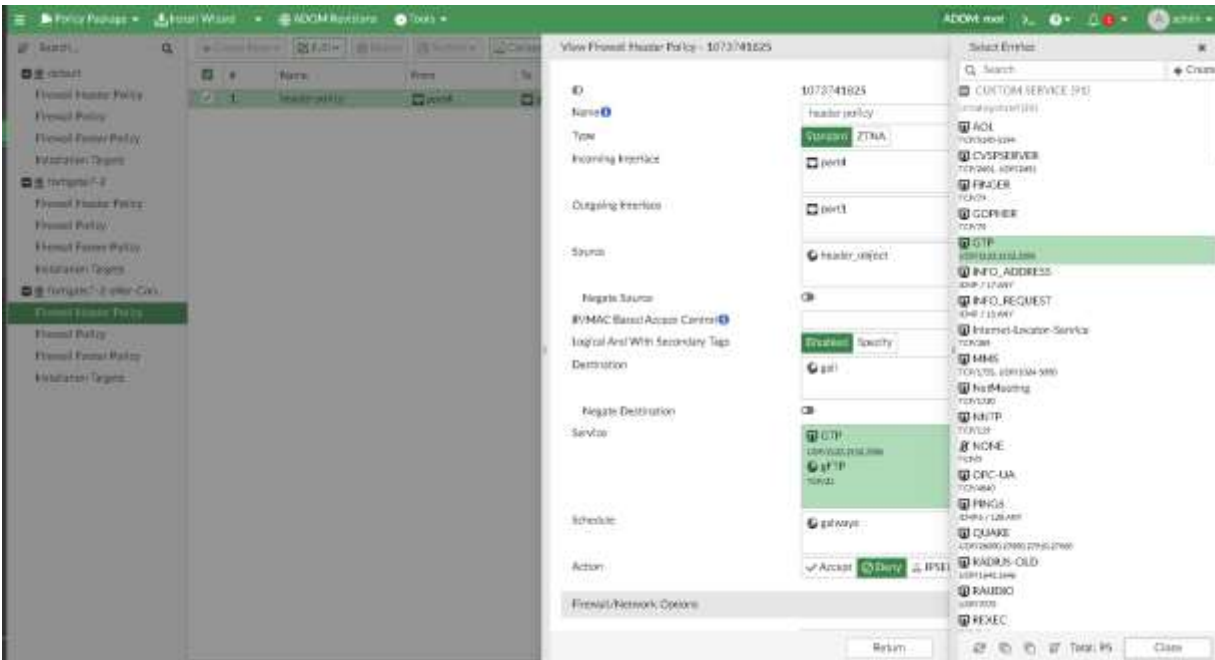


Figure 14 — View Firewall Header Policy — adding the GTP service is offered but not accepted.



Figure 15 — The header policy unchanged after the attempt.

This is the control working as designed

An ADOM administrator can see the global rules and cannot alter them. Changing a header or footer policy means going back to the Global Database and reassigning.

It is also the behaviour to recognise when someone reports that a policy "will not save" — the question to ask is which ADOM owns it, not what is wrong with the package.

4.2 The objects came down too

Object Name	Address	IP Netmask	Any	Admin / 2026-10-07 11:07:56 PDT
route	Address	IP Netmask: 0.0.0.0/255.255.255	any	admin / 2026-10-07 11:07:56 PDT
login.microsoftonline.com	Address	FQDN:login.microsoftonline.com	any	admin / 2026-10-07 11:07:56 PDT
login.microsoft.com	Address	FQDN:login.microsoft.com	any	admin / 2026-10-07 11:07:56 PDT
login.windows.net	Address	FQDN:login.windows.net	any	admin / 2026-10-07 11:07:56 PDT
gmail.com	Address	FQDN:gmail.com	any	admin / 2026-10-07 11:07:56 PDT
wildcard.google.com	Address	FQDN:*google.com	any	admin / 2026-10-07 11:07:56 PDT
wildcard.dropbox.com	Address	FQDN:*dropbox.com	any	admin / 2026-10-07 11:07:56 PDT
SSLVPN_TUNNEL_ADDR1	Address	IP Range: 10.212.134.200-10.212.134.210	any	admin / 2026-10-07 11:07:56 PDT
all	Address	IP Netmask: 0.0.0.0/0.0.0.0	any	admin / 2026-10-07 11:07:56 PDT
FIREWALL_AUTH_PORTAL_ADDRESSES	Address	IP Netmask: 0.0.0.0/0.0.0.0	any	admin / 2026-10-07 11:07:56 PDT
FABRIC_DEVICE	Address	IP Netmask: 0.0.0.0/0.0.0.0	any	IPv4 addresses of Fabric Devi admin / 2026-10-07 11:07:56 PDT
metasploit-server	Address	IP Netmask: 169.254.169.254/255.255.255.255	any	admin / 2026-10-07 11:07:56 PDT
RFC1918-10	Address	IP Netmask: 10.0.0.0/255.0.0.0	any	admin / 2026-10-07 11:07:56 PDT
RFC1918-172	Address	IP Netmask: 172.16.0.0/255.240.0.0	any	admin / 2026-10-07 11:07:56 PDT
RFC1918-192	Address	IP Netmask: 192.168.0.0/255.255.0.0	any	admin / 2026-10-07 11:07:56 PDT
LAN-Net	Dynamic Address	IP Netmask: 192.168.10.0/255.255.255.0	any	admin / 2026-10-08 09:55:13 PDT
Site-LAN	Address	IP Netmask: 10.38.220.0/255.255.255.0	any	admin / 2026-10-08 10:30:17 PDT
header_object	Address	IP Netmask: 172.16.10.0/255.255.255.0	any	admin / 2026-10-08 20:27:52 PDT
gall	Address	IP Netmask: 0.0.0.0/0.0.0.0	any	admin / 2026-10-08 20:27:52 PDT

Figure 16 — The root ADOM address list — header_object and gall now present alongside the local objects.

Because the assignment used USED objects only, the ADOM received header_object, gall and the services and schedule the two policies reference, and nothing else from the global object table. They sit in the same list as LAN-Net and Site-LAN from Labs 06 and 07, which is why the naming convention matters.

05. FortiGuard Distribution

A FortiManager can act as the FortiGuard distribution point for the devices it manages, downloading antivirus, IPS and filtering packages once and serving them locally rather than having every firewall reach the internet.



Figure 17 — FortiGuard Server and Service Settings.

Setting	What it controls
Communication with FortiGuard Server	Global Servers or US-only. A jurisdiction choice as much as a performance one.
Enable AntiVirus and IPS Service	Which product families and firmware versions FortiManager downloads packages for. Each version ticked is storage and bandwidth spent.
Server Override Mode	Strict restricts devices to the override server; Loose allows them to fall back to other FortiGuard servers if it is unreachable.

Observation	No FortiGate version is ticked in the AntiVirus and IPS list, and the Web Filter and Email Filter services are off.
Effect	FortiManager is not downloading any package for the managed FortiGate, so it cannot serve one. Devices pointed at it for updates would get nothing.
Action	Tick 7.2 to match the managed device before relying on local distribution. In the lab this is harmless because the VMs are unlicensed for FortiGuard anyway.

► 5.1 Download prioritisation

Where bandwidth is constrained, FortiManager can be told which products to fetch first.

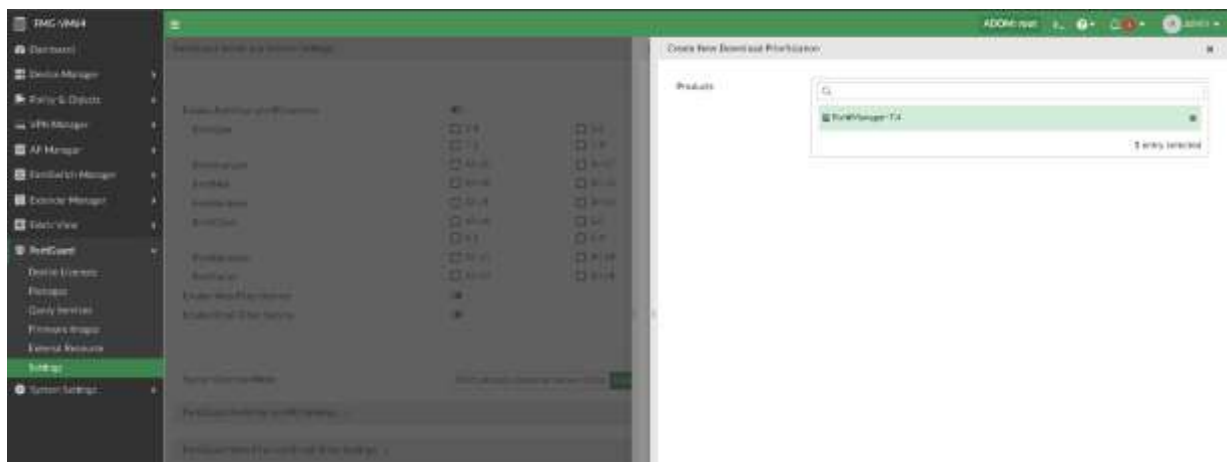


Figure 18 — Create New Download Prioritization — ordering by product.

Enable By Package ☐

+ Create New Delete Move To

☐	#	Package Name	Product	Version	Service Entitlement	Type	Latest Version (Release Date/Time)	Size	
No record found.									
									0

Figure 19 — Prioritisation can also be set per package rather than per product.

06. Firmware Templates

A firmware template defines a target version per platform and an install window, and is assigned to devices or groups in the same way as the provisioning templates in Lab 06.

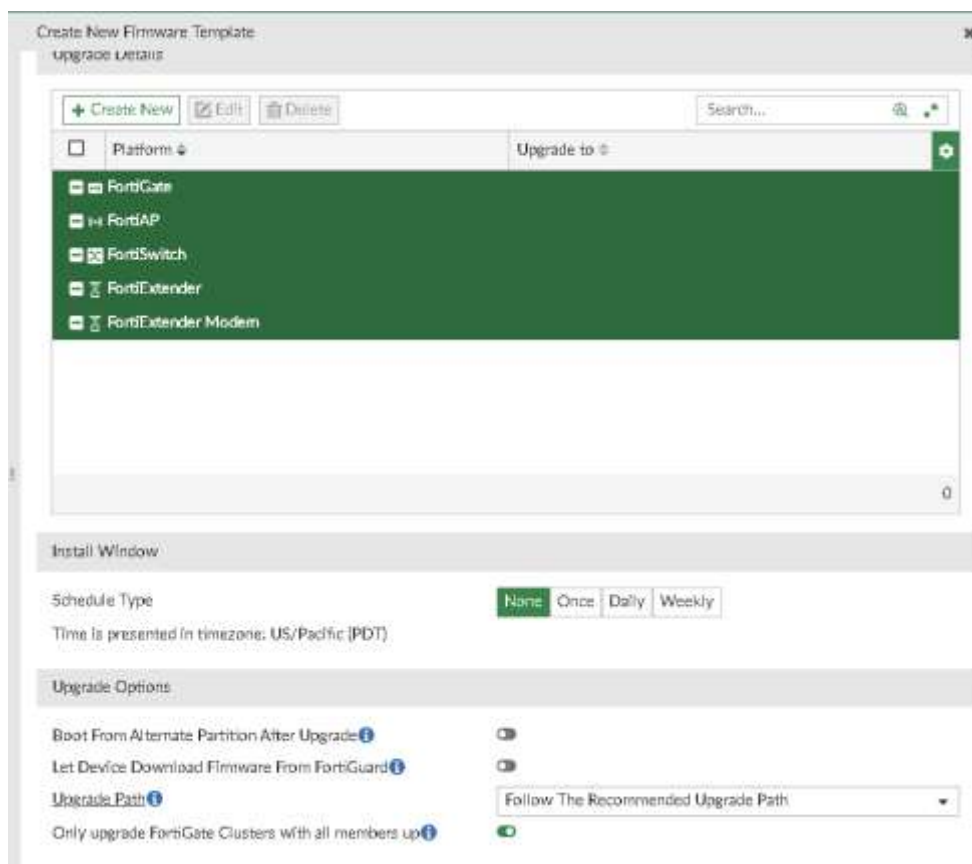


Figure 20 — Create New Firmware Template — per-platform target versions, install window and upgrade options.

Option	What it does
Schedule Type	None, Once, Daily or Weekly. The install window is when the upgrade is allowed to run, not when it is queued.
Boot From Alternate Partition After Upgrade	Leaves the previous image bootable — the rollback path if the new version misbehaves.
Let Device Download Firmware From FortiGuard	Off means FortiManager serves the image; on means each device fetches it itself.
Follow The Recommended Upgrade Path	FortiManager steps through the intermediate versions Fortinet requires rather than jumping straight to the target.
Only upgrade FortiGate Clusters with all members up	Prevents an HA upgrade starting while a member is already down.

The recommended upgrade path is not optional in practice

Jumping several major versions in one step is unsupported and routinely corrupts configuration. Leaving this setting on is what turns a firmware template from a convenience into a safe one.

► 6.1 Assigning the template

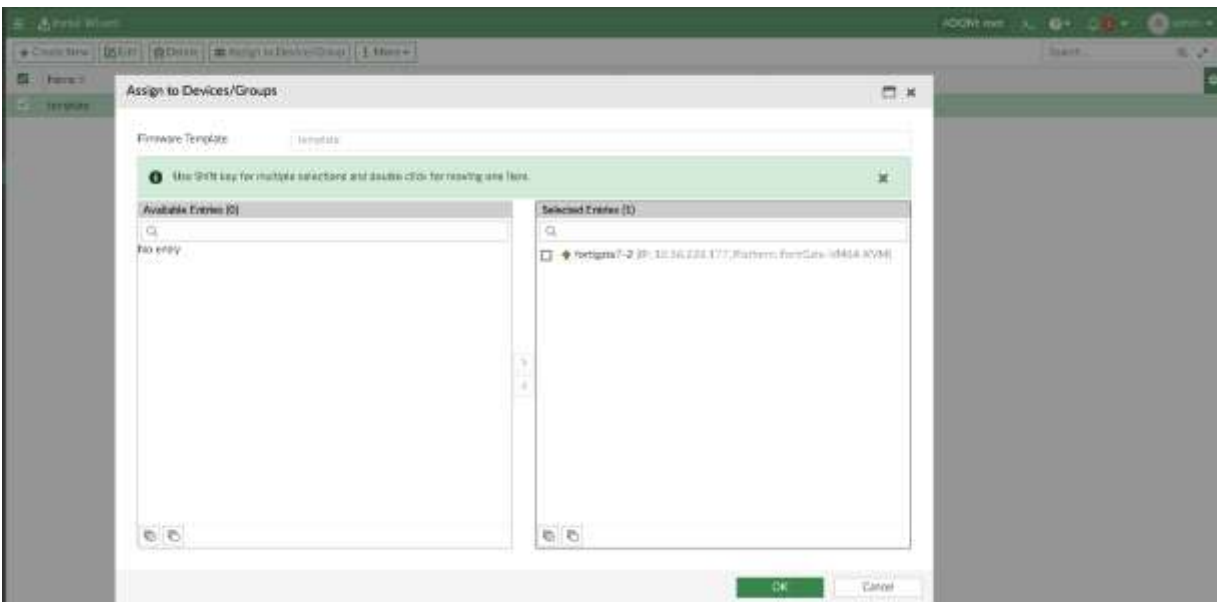


Figure 21 — Assign to Devices/Groups — fortigate7-2 selected.



Figure 22 — The template assigned, with no schedule set.

► 6.2 Vulnerable devices

FortiManager cross-references the firmware running on managed devices against Fortinet's published advisories, and the result is the practical reason firmware templates exist.

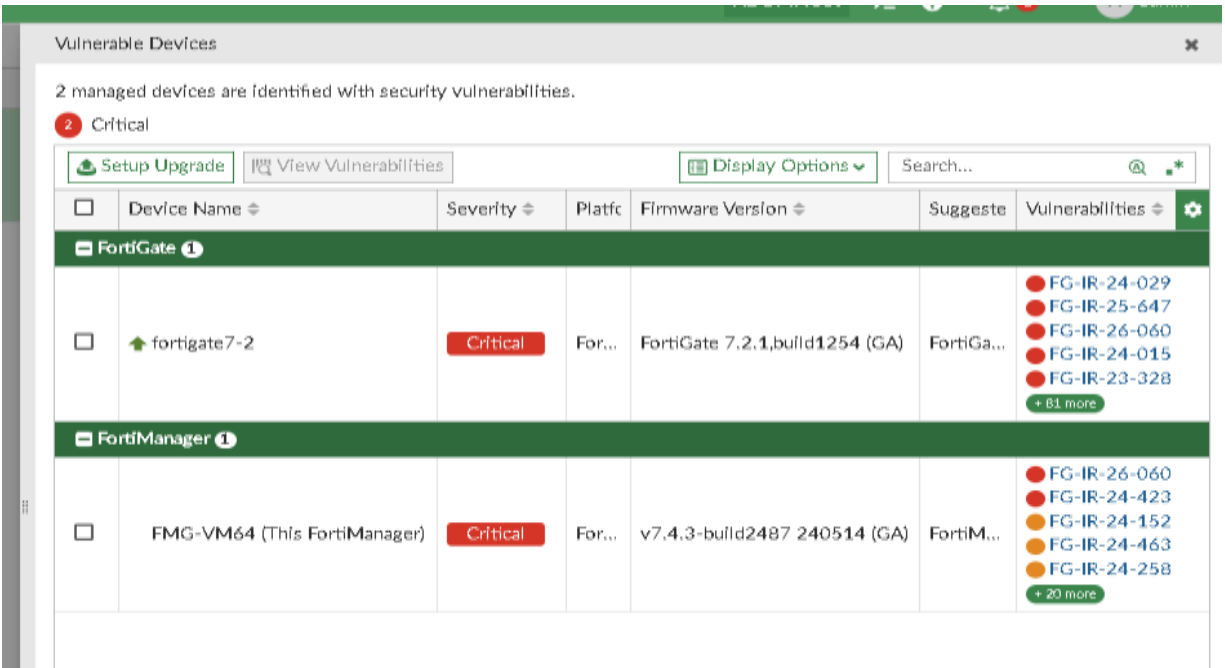


Figure 23 — Vulnerable Devices — both appliances flagged Critical.

Observation	Both the FortiGate on 7.2.1 and FortiManager itself on 7.4.3 are listed Critical, with 86 and 25 advisories respectively.
Why	The pair is pinned to these versions because that combination is what allowed FGFM registration in Lab 04. It is a lab constraint, not a supportable configuration.
Note	Worth stating wherever this lab set is published, so the version pairing is not read as a recommendation. On a real estate this screen is the input to the upgrade plan.

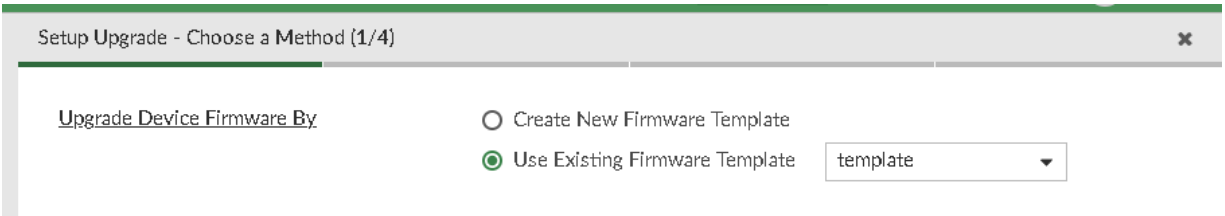


Figure 24 — Setup Upgrade — the vulnerability view feeding straight into the firmware template.

The Setup Upgrade control on that screen builds the upgrade from either a new template or the existing one, which closes the loop: the vulnerability report identifies the devices, and the